Multirate Stein Variational Gradient Descent for Efficient Bayesian Sampling

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Abstract

Many particle-based Bayesian inference methods use a single global step size for all parts of the update. In Stein variational gradient descent (SVGD), however, each update combines two qualitatively different effects: attraction toward high-posterior regions and repulsion that preserves particle diversity. These effects can evolve at different rates, especially in high-dimensional, anisotropic, or hierarchical posteriors, so one step size can be unstable in some regions and inefficient in others. In response, we derive a multirate version of SVGD that updates these components on different time scales. The framework yields practical algorithms, including a symmetric split method, a fixed multirate method (MR-SVGD), and an adaptive multirate method (Adapt-MR-SVGD) with local error control. We evaluate the methods in a broad and rigorous benchmark suite covering six problem families: a 50D Gaussian target, multiple 2D synthetic targets, UCI Bayesian logistic regression, multimodal Gaussian mixtures, Bayesian neural networks, and large-scale hierarchical logistic regression. Evaluation combines distributional fidelity, predictive performance, calibration, mixing, and explicit computational cost accounting. Across this diverse benchmark suite, multirate SVGD variants show improved robustness and more favorable quality-cost tradeoffs than vanilla SVGD, with the strongest gains on difficult hierarchical and strongly anisotropic targets.

Keywords: Stein variational gradient descent, multirate integration, particle methods, kernel Stein discrepancy

1. Introduction

Modern Bayesian inference often requires efficient sampling from complex, high-dimensional, and multimodal posterior distributions. As these problems grow in scale and complexity, traditional sampling methods face significant computational and statistical challenges. The advent of automatic differentiation has enabled the widespread use of gradient-based samplers, making methods such as stochastic gradient Langevin dynamics (SGLD) and stochastic gradient Hamiltonian Monte Carlo (SGHMC) practical and popular choices for large-scale applications [1, 2]. While these approaches advance a single Markov chain using noisy gradients, an alternative paradigm is offered by interacting particle methods, which evolve a population of particles whose empirical distribution approximates the target posterior. This population-based view is central to particle-based variational inference (ParVI), which interprets interacting particle systems as finite-particle transport approximations on probability measures [3]. Within this class, SVGD is a prominent deterministic representative, and recent work has examined acceleration, particle degeneracy, and kernel design in related Stein-particle methods [3, 4, 5]. It opens new avenues for improving sampling efficiency and robustness, especially in challenging inference settings.

A useful unifying perspective is to view many such methods as discretizations

of a particle flow (possibly with a stochastic component) that transports an initial set of particles toward the target distribution. In this work, we focus on Stein variational gradient descent (SVGD) [6] as a representative interacting particle method. SVGD constructs a velocity field for the particles based on the target density and a reproducing kernel Hilbert space (RKHS) structure, and updates the particles by discretizing this flow. The explicit form of the SVGD update and its velocity field are given in eqs. (1) and (2) in Section 2. The velocity field naturally separates into an attractive drift (moving particles toward high-density regions) and a repulsive interaction (spreading particles to maintain diversity and support coverage of multiple modes).

Despite its conceptual simplicity, SVGD presents a numerical challenge that is central to this paper. The drift and repulsion components can impose different stability constraints on the discretization: when particles crowd, repulsion can become stiff and destabilize updates, while on anisotropic targets the drift can require finer resolution to faithfully follow the geometry of $\log p$.

A single step size must therefore balance competing requirements and can either over-damp repulsion (leading to loss of diversity and mode coverage) or under-resolve drift (leading to slow progress). Moreover, repulsion typically requires $O(N^2)$ kernel evaluation per step, so increasing stability by oversampling repulsion can become computationally expensive. These numerical tensions interact with a known statistical pathology of SVGD-type methods: repulsive forces can weaken in high dimensions or under limited particle resolution, leading to particle collapse and poor support of multiple modes [4, 7]. In practice, the resulting single-step-size compromise is both statistically and computationally inefficient: it can degrade support coverage and predictive quality while still spending substantial kernel work to stabilize the wrong part of the update.

A common approach to solving systems with significantly different components is to use partitioned methods such as implicit-explicit (IMEX) [8, 9] and multirate integration methods [10]. In the context of ordinary differential equations (ODEs), multirate methods exploit a natural splitting of the dynamics into fast and slow components, allowing each to be integrated with a step size appropriate to its scale [11, 12, 13]. For example, in a split ODE of the form $\dot{x} = \phi_{\text{fast}}(x) + \phi_{\text{slow}}(x)$, multirate schemes apply more frequent, smaller steps to the stiff or rapidly varying part, while updating the slow part less often. This approach improves both stability and efficiency in multi-scale dynamical systems, and has been successfully applied in a variety of scientific computing contexts.

The structure of the SVGD update naturally separates into two forces: an attractive drift and a repulsive interaction. These forces often operate on different time scales and can impose distinct numerical challenges. Motivated by advances in multirate time-integration for multi-scale dynamical systems—including MrGARK, IMEX, and related splitting schemes [11, 12, 13, 9, 14]—we propose to integrate the drift and repulsion terms on separate time scales. This leads to a family of multirate SVGD variants, including symmetric splitting, fixed repulsion substepping, and adaptive error-controlled drift substepping.

To assess the effectiveness of these methods, we conduct a comprehensive empirical evaluation. We compare our multirate SVGD variants with established approaches from the literature across a range of benchmark problems, using a variety of task-appropriate metrics. Results highlight the strengths and limitations of each method under diverse geometric and statistical challenges.

1.1. Contributions

Our main contributions are:

- Multirate SVGD formulations that explicitly separate drift and repulsion and support symmetric splitting and fixed multirate schedules.
- An adaptive error-controlled multirate SVGD variant that selects drift substeps based on a local error estimator while keeping repulsion stable.
- A benchmark suite spanning 50D, 2D, UCI, mixture, BNN, and HLR targets, with task-appropriate diagnostics, early stopping, and gradient/kernel evaluation accounting for efficiency comparisons.

Organization.. Section 2 presents the SVGD variants and the multirate construction. Section 3 describes the benchmarks and experimental protocol and reports results. Section 4 summarizes findings and discusses directions for future work.

2. Method

2.1. Background: SVGD as a particle flow

Let $p(x)$ denote a target density on $\mathbb{R}^d$ and let $\{x_i^t\}_{i=1}^N$ be a set of particles at iteration t. Stein variational gradient descent (SVGD) evolves particles by following a deterministic interacting-particle flow whose velocity field is constrained to a reproducing kernel Hilbert space (RKHS) [6]. A first-order discretization of this flow is the update

$$x_i^{t+1} = x_i^t + h\,\phi(x_i^t), \tag{1}$$

where $h > 0$ is a step size and the SVGD velocity field is

$$\phi(x) = \frac{1}{N}\sum_{j=1}^{N}\Big[k(x_j, x)\,\nabla_{x_j}\log p(x_j) + \nabla_{x_j}k(x_j, x)\Big]. \tag{2}$$

The first term in eq. (2) is an *attractive* drift that pushes particles toward high-density regions via the score $\nabla \log p$. The second term is a *repulsive* interaction that discourages particle collapse.

For derivation and notation, we write the method with a single RBF kernel

$$k(x, y) = \exp\Big(-\frac{\|x-y\|_2^2}{2\,h_k}\Big), \tag{3}$$

where the bandwidth h_k is chosen by the median heuristic (the median pairwise squared distance among particles) and scaled by a factor $\alpha > 0$ to modulate repulsion strength. Most experiments use this single-kernel form. For the Mix8 multimodal benchmark we additionally consider a shared multiscale RBF variant to test whether broader particle interactions improve mode exploration; the drift–repulsion splitting and multirate updates are otherwise unchanged. More generally, once the attractive and repulsive fields are defined, the multirate constructions below are agnostic to the particular admissible kernel family used to build those fields.

2.2. Drift–repulsion splitting and multirate integration

The central numerical observation motivating this work is that the drift and repulsion components of eq. (2) can impose different stability constraints and can naturally evolve on different time scales. We therefore write the SVGD flow as a split system

$$\dot{x} = f_{\mathrm{rep}}(x) + f_{\mathrm{drift}}(x), \tag{4}$$

where, for particle i,

$$f_{\mathrm{drift}}(x_i) := \frac{1}{N}\sum_{j=1}^{N} k(x_j, x_i)\,\nabla_{x_j} \log p(x_j), \tag{5}$$

$$f_{\mathrm{rep}}(x_i) := \frac{1}{N}\sum_{j=1}^{N} \nabla_{x_j} k(x_j, x_i). \tag{6}$$

We next define multirate discretizations that apply different substepping strategies to f_{drift} and f_{rep} while keeping a common macro-step size h.

Strang-split SVGD. Define the forward Euler maps that advance the split system by a step size τ:

$$\Psi^{\tau}_{\mathrm{rep}}(x) = x + \tau f_{\mathrm{rep}}(x), \qquad \Psi^{\tau}_{\mathrm{drift}}(x) = x + \tau f_{\mathrm{drift}}(x).$$

Strang splitting [15] is a classical second-order method that constructs a macro-step of size h by composing such substeps. In our setting, one macro step takes the form

$$x^{t+1} = \Psi^{h/2}_{\mathrm{rep}} \circ \Psi^{h}_{\mathrm{drift}} \circ \Psi^{h/2}_{\mathrm{rep}}(x^t), \tag{7}$$

which alternates half a repulsion step, a full drift step, and another half repulsion step. In practice, the half-steps can be implemented as $\frac{h}{2m}$ smaller substeps for stability.

Algorithm 1 Fixed multirate SVGD (MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h, repulsion substeps m, kernel k

$x \leftarrow \{x_i\}$
for $s = 1$ to m **do**
  compute repulsion $f_{\text{rep}}(x)$ using eq. (6)
  $x \leftarrow x + (h/m)\, f_{\text{rep}}(x)$
end for
compute drift $f_{\text{drift}}(x)$ using eq. (5)
$x \leftarrow x + h\, f_{\text{drift}}(x)$
return x

Fixed multirate SVGD (MR-SVGD). Fixed multirate integrators provide a principled way to allocate different step sizes to fast and slow components of a split system [10, 11, 12]. In our SVGD setting, however, higher-order multirate schemes are often not cost-effective: their additional stages translate into multiple evaluations of the drift and repulsion fields per macro step, and each such evaluation is expensive due to gradient computations and, in particular, $O(N^2)$ kernel interactions. Moreover, practical accuracy is also limited by finite-particle effects and by choices such as bandwidth selection and early stopping, which can mute the benefits of increasing time-integration order at a fixed computational budget. We therefore specialize to a first-order multirate construction (Euler update maps) that preserves a macro step of size h for the drift while resolving the repulsion with m substeps:

$$x^{t+1} = \Psi_{\text{drift}}^{h} \circ \left(\Psi_{\text{rep}}^{h/m}\right)^{m} (x^t). \tag{8}$$

This retains a coarse drift step while stabilizing the repulsive interaction when particles crowd, at the cost of additional kernel evaluations. The corresponding implementation-level procedure is summarized in algorithm 1.

Adaptive multirate SVGD (Adapt-MR-SVGD). Although fixed-m schedules are simple and robust, they impose a uniform drift resolution along the entire trajectory. As a result, they may spend unnecessary computational effort in benign regions while providing insufficient resolution in locally stiff regimes. To address this imbalance, we choose the number of drift microsteps m adaptively at each macro step, while retaining one repulsion update per macro step. This follows the standard local-error-control perspective used in ODE integrators, where local error estimates are used to adjust resolution [16]. After the repulsion update, denote the state by $x = \Psi^h_{\mathrm{rep}}(x^t)$. We estimate the local drift-discretization error by comparing one drift step of size h with two consecutive drift half-steps of size $h/2$,

$$x_{\mathrm{full}} = x + h\, f_{\mathrm{drift}}(x), \tag{9}$$

$$x_{\mathrm{half}} = x + \tfrac{h}{2}\, f_{\mathrm{drift}}(x), \tag{10}$$

$$\tilde{x}_{\mathrm{full}} = x_{\mathrm{half}} + \tfrac{h}{2}\, f_{\mathrm{drift}}(x_{\mathrm{half}}). \tag{11}$$

We then form the relative discrepancy

$$\rho = \frac{\|\tilde{x}_{\mathrm{full}} - x_{\mathrm{full}}\|_2}{\|x\|_2 + \epsilon}. \tag{12}$$

which serves as a dimensionless local error indicator for the drift update. For a first-order drift integrator, the step-doubling discrepancy behaves locally as

$$\rho(h) = Ch^2 + \mathcal{O}(h^3).$$

If the drift step is replaced by m microsteps of size h/m, then

$$\rho_m \approx C(h/m)^2 \approx \rho/m^2.$$

Imposing the target local accuracy $\rho_m \approx \varepsilon$ gives $m \approx \sqrt{\rho/\varepsilon}$. We therefore use the clipped integer controller

$$m \leftarrow \mathrm{clip}\Big(\Big\lceil \sqrt{\rho/\varepsilon}\Big\rceil, m_{\min}, m_{\max}\Big), \tag{13}$$

which increases m when $\rho > \varepsilon$ and decreases it when $\rho \leq \varepsilon$, while enforcing the robustness bounds $m_{\min} \leq m \leq m_{\max}$. The macro-step update is then

$$x^{t+1} = \Big(\Psi^{h/m}_{\mathrm{drift}}\Big)^m \circ \Psi^h_{\mathrm{rep}}(x^t), \qquad m = m(x^t, h). \tag{14}$$

Here $\varepsilon > 0$ is a user-set tolerance and $\epsilon > 0$ is a small constant to avoid division by zero. This strategy increases drift resolution only when the local error indicator suggests stiffness or rapid variation in the drift component. See algorithm 2 for complete implementation.

Remark 1. In implementation, if $m \leq 2$ we reuse the already computed step-doubling trajectory to avoid extra drift evaluations.

Algorithm 2 Adaptive multirate SVGD (Adapt-MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h, bounds $m_{\min}, m_{\max}$, tolerance ε

$x \leftarrow \{x_i\}$
compute repulsion $f_{\text{rep}}(x)$ using eq. (6)
$x \leftarrow x + h\, f_{\text{rep}}(x)$
compute drift $f_{\text{drift}}(x)$ using eq. (5)
$x_{\text{full}} \leftarrow x + h\, f_{\text{drift}}(x)$
$x_{\text{half}} \leftarrow x + (h/2)\, f_{\text{drift}}(x)$
compute drift $f_{\text{drift}}(x_{\text{half}})$
$\tilde{x}_{\text{full}} \leftarrow x_{\text{half}} + (h/2)\, f_{\text{drift}}(x_{\text{half}})$
estimate relative error $\rho = \|\tilde{x}_{\text{full}} - x_{\text{full}}\| / (\|x\| + \epsilon)$
$m \leftarrow \text{clip}(\lceil \sqrt{\rho/\varepsilon} \rceil, m_{\min}, m_{\max})$
if $m \leq 2$ **then**
  $x \leftarrow \tilde{x}_{\text{full}}$ ▷ reuse step-doubling path
else
  for $s = 1$ to m **do**
    compute drift $f_{\text{drift}}(x)$
    $x \leftarrow x + (h/m)\, f_{\text{drift}}(x)$
  end for
end if
return x

Numerical guards. To improve robustness, we clip gradients and clamp extreme state values before and after the split updates, and we replace non-finite updates with the last finite state. These safeguards are implementation-level stability controls and do not alter the intended fixed point of the discretization.

3. Experiments and Results

We evaluate multirate SVGD variants on six benchmark groups: a 50D Gaussian target, a suite of 2D synthetic targets, UCI logistic regression, a 2D Gaussian mixture (mix8), a one-hidden-layer Bayesian neural network (BNN), and a large-scale hierarchical logistic-regression (HLR) benchmark. The comparison focuses on both statistical quality and computational cost.

3.1. Evaluation Metrics

Let $X = \{x_i\}_{i=1}^N$ denote the checkpoint sample set (particles for particle methods; a rolling window for single-chain methods). We use complementary diagnostics that separately assess distributional fidelity, predictive quality, and computational cost.

Moment fidelity. When reference moments are available, we report $e_\mu = \|\hat{\mu} - \mu_{\mathrm{ref}}\|_2$ and $e_\Sigma = \|\hat{\Sigma} - \Sigma_{\mathrm{ref}}\|_F$, where $\hat{\mu} = \frac{1}{N}\sum_{i=1}^N x_i$ and $\hat{\Sigma} = \frac{1}{N}\sum_{i=1}^N (x_i - \hat{\mu})(x_i - \hat{\mu})^\top$. These summarize first- and second-order bias, which is critical in the Gaussian and synthetic-target settings.

Stein discrepancy. For distributional fidelity, we use kernel Stein discrepancy (KSD) with score $s_p(x) = \nabla_x \log p(x)$ and an RBF base kernel $k_h(\cdot, \cdot)$ [17, 18]. The corresponding Stein kernel is

$$\begin{aligned} \kappa_p(x, x') = {} & s_p(x)^\top s_p(x')\, k_h(x, x') + s_p(x)^\top \nabla_{x'} k_h(x, x') \\ & + s_p(x')^\top \nabla_x k_h(x, x') + \mathrm{tr}(\nabla_x \nabla_{x'} k_h(x, x'))\,, \end{aligned} \tag{15}$$

and the empirical discrepancy is

$$\mathrm{KSD}(X; p) = \left(\frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \kappa_p(x_i, x_j) \right)^{1/2}. \tag{16}$$

Under standard kernel conditions, KSD is zero only at the target law and increases under global mismatch, including location bias, covariance distortion, and mode omission [17, 18].

Mean log-density fit. We complement KSD with the empirical mean log-density

$$\overline{\log p}(X) = \frac{1}{N} \sum_{i=1}^N \log p(x_i), \tag{17}$$

which corresponds to the logarithmic scoring rule [19]. This metric quantifies average concentration in high-density regions, but it is local in $\log p(x)$ and therefore does not, by itself, certify global distributional agreement. Reporting eqs. (16) and (17) separates global fit from local density concentration.

Predictive performance and calibration. For UCI and BNN tasks, we report accuracy, negative log-likelihood NLL $= -\frac{1}{M}\sum_{m=1}^{M}\big[y_m \log \hat{p}_m + (1 - y_m)\log(1-\hat{p}_m)\big]$, and expected calibration error ECE $= \sum_{b=1}^{B} \frac{|\mathcal{B}_b|}{M} |\mathrm{acc}(\mathcal{B}_b) - \mathrm{conf}(\mathcal{B}_b)|$. This combination separates discrimination (accuracy), probabilistic sharpness (NLL), and calibration quality (ECE).

Mode-structure diagnostics. For structured multimodal targets, we report mode coverage, normalized mode entropy, and mode imbalance to detect mode collapse and uneven mass allocation; minimum mode mass is retained as a secondary check when informative. For the 2D synthetic targets, we additionally retain grid L_1 discrepancy and moment diagnostics as secondary spatial mismatch measures.

Cost accounting. In addition to wall-clock time, we track gradient- and kernel-evaluation counts to support hardware-agnostic efficiency comparisons across methods with different per-step workloads.

Mixing efficiency. To quantify dependence, we report ESS from a one-dimensional diagnostic sequence $z_1, \ldots, z_T$ built from the first coordinate in our implementation. For particle methods at a checkpoint, z_i is the first coordinate of particle i ($i = 1, \ldots, N$). For single-chain baselines, z_i is the first coordinate of the i-th state in a rolling window of recent iterates (window length as set in each benchmark script). Define

$$\bar{z} = \frac{1}{T}\sum_{i=1}^{T} z_i, \qquad \hat{\gamma}_k = \frac{1}{T-k}\sum_{i=1}^{T-k}(z_i - \bar{z})(z_{i+k} - \bar{z}), \qquad \hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}. \tag{18}$$

Autocorrelations are evaluated up to $k_{\max} = \min(\lfloor T/2 \rfloor, 100)$. Let K be the first lag in $\{1, \ldots, k_{\max}\}$ such that $\hat{\rho}_K < 0$; if no such lag exists, we set $K = k_{\max}$. The reported ESS is

$$\mathrm{ESS}(z) = \frac{T}{1 + 2\sum_{k=1}^{K} \hat{\rho}_k}. \tag{19}$$

This initial-positive truncation controls high-lag noise in the empirical autocorrelation and provides a stable scalar proxy for mixing. Because eq. (19) is computed from a single diagnostic coordinate, we interpret ESS as a univariate mixing indicator rather than a full multivariate efficiency estimate. Accordingly, distribution-level conclusions rely primarily on KSD and moment/mode diagnostics, with ESS used as a complementary statistic.

Lower is better for e_μ, e_Σ, KSD, NLL, ECE, and grid L_1, whereas higher is better for ESS, accuracy, coverage, entropy, minimum mode mass, and mean log-probability.

3.2. Experimental protocol

Across benchmarks, we compare the same method family whenever applicable: vanilla SVGD, Strang-split SVGD, fixed multirate SVGD (MR-SVGD), adaptive multirate SVGD (Adapt-MR-SVGD), and the single-chain baselines SGLD and an SGHMC-style method. The SVGD-family updates are those defined in section 2, namely eqs. (1), (7), (8) and (14).

Remark 2. For the single-chain baselines, we use the standard stochastic-gradient update

$$x_{t+1} = x_t + \eta \nabla \log p(x_t) + \sqrt{2\eta}\,\xi_t, \qquad \xi_t \sim \mathcal{N}(0, I), \tag{20}$$

for SGLD [1], and

$$v_{t+1} = \beta v_t + \eta \nabla \log p(x_t) + \sqrt{2\alpha\eta}\,\xi_t, \qquad x_{t+1} = x_t + v_{t+1}, \tag{21}$$

for an SGHMC-style baseline [2]. Here η is the step size, α is a friction/noise parameter, and $\beta \in (0, 1)$ is momentum decay.

For particle methods, we use $N = 128$ particles; for single-chain methods, we compute metrics from a rolling window of recent iterates to obtain comparable finite-sample summaries. Alongside wall-clock time, we report gradient- and kernel-evaluation counts as hardware-agnostic measures of computational cost.

All benchmarks use checkpointed evaluation at a benchmark-specific interval and a patience-based early-stopping rule triggered by persistent degradation of a designated monitor metric. Pure sampling benchmarks use KSD as the monitor (50D Gaussian, 2D synthetic targets, and Mixture2D), whereas predictive benchmarks use held-out NLL (UCI logistic regression, BNN, and HLR). For each run, we retain the best finite checkpoint under that monitor before stopping. Accuracy and ECE are reported for predictive tasks but are never used as stopping triggers. Benchmarks with pronounced numerical instability add a short non-finite fail-fast guard; we note those cases locally.

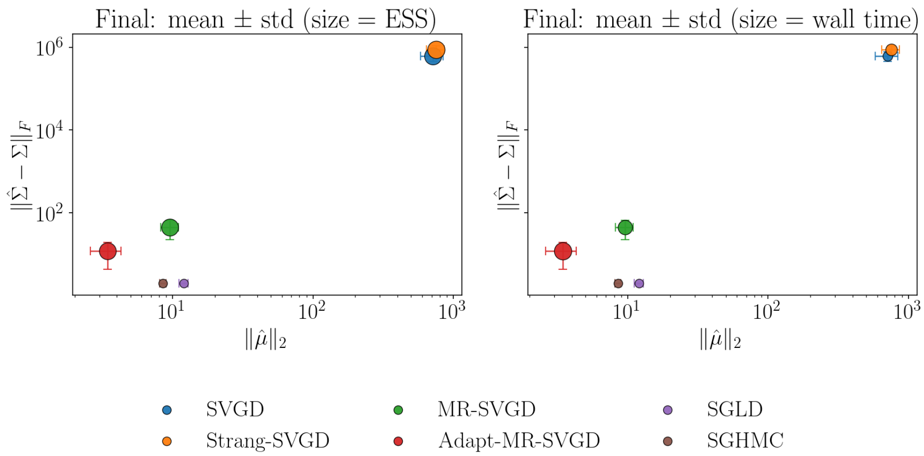

Figure 1: 50D Gaussian: final mean ± std Pareto plot in moment-error space with marker size encoding ESS (left) and wall time (right).

3.3. 50D Gaussian

This benchmark stresses stability in a high-dimensional setting with anisotropic covariance structure. We run 1,000 iterations with checkpointing every 50 steps and a 5-seed ablation. Reported metrics are moment errors ($\|\hat{\mu} - \mu\|_2$ and $\|\hat{\Sigma} - \Sigma\|_F$), ESS, KSD, and mean log probability.

Among the particle methods, Adapt-MR-SVGD is the only variant that keeps both moment errors and KSD under control in this anisotropic Gaussian setting: it substantially improves on MR-SVGD and avoids the severe degradation exhibited by vanilla SVGD and Strang-SVGD. The chain baselines remain competitive on some final-checkpoint fidelity metrics, but they do so with ESS near 4, far below the particle methods. The main lesson of this benchmark is therefore numerical rather than absolute: multirate treatment is critical for maintaining stable, informative particle behavior once drift and repulsion interact through strong anisotropy.

3.4. 2D synthetic targets

We evaluate banana, ring, squiggly, two-moons, and funnel targets over five seeds, with 1,000 iterations and checkpointing every 20 steps. Here KSD serves both as the stop monitor and the primary quality metric, and we add a short non-finite KSD fail-fast guard because several targets can destabilize weak particle updates. We report mean log probability as a secondary density-concentration score, gradient/kernel evaluations and wall time as cost

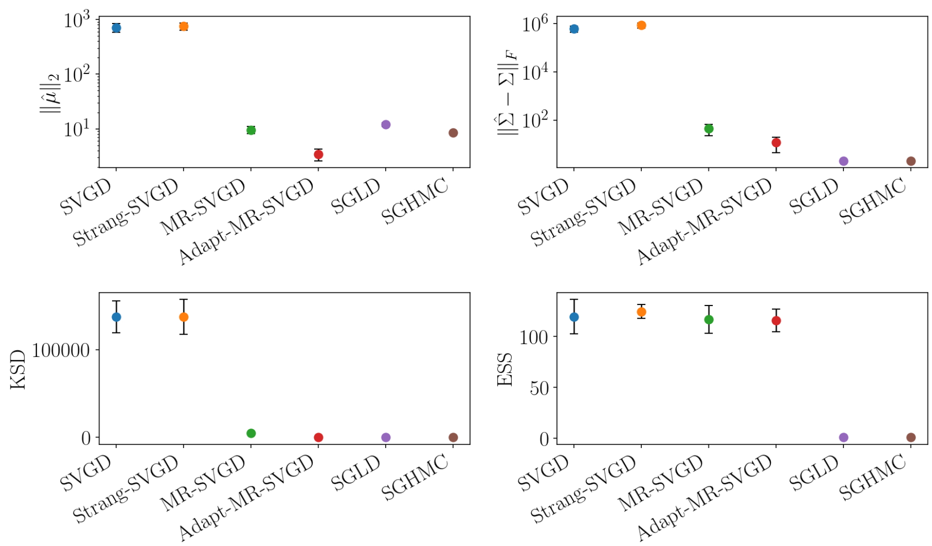

Figure 2: 50D Gaussian final-checkpoint summary across methods. The four panels report $\|\hat{\mu} - \mu\|_2$, $\|\hat{\Sigma} - \Sigma\|_F$, KSD, and ESS, respectively; markers show mean values across seeds and error bars indicate one standard deviation. Lower is better for the first three metrics, whereas higher is better for ESS.

metrics, and ESS as a supplemental mixing proxy. Moment errors and grid-based L_1 discrepancy are retained as diagnostics, but they are not used in the main 2D summary table. To contextualize the geometry of these targets, we also show short-run visualization panels for representative cases.

Across the full 2D suite, Adapt-MR-SVGD gives the strongest overall distributional fidelity: it is the only particle method with zero failures, attains the best overall median KSD, and also attains the best median mean log-density while retaining high ESS. The gain is concentrated on the geometry-driven ring, squiggly, and two-moons targets, where Adapt-MR-SVGD achieves the best mean KSD over seeds (approximately 0.231, 1.03, and 0.308, respectively) and the single-rate particle methods degrade sharply. On the easier banana target, by contrast, all particle methods attain nearly identical KSD values around 0.116, indicating that multirate treatment is not especially beneficial when the target geometry is comparatively mild.

The funnel target remains the main exception. Fixed MR-SVGD attains the best mean KSD on funnel (0.261 versus 0.456 for Adapt-MR-SVGD), but Adapt-MR-SVGD still remains fully stable and yields substantially better mean log-density (-0.023 versus -2.89). Vanilla SVGD and Strang-

Method	Fails↓	KSD↓	mean logp↑	ESS↑	grad evals↓	wall (s)↓
Adapt-MR-SVGD	0	0.308	-0.175	128.0	182	0.79
MR-SVGD	0	40.13	-278.0	121.8	80	1.13
SVGD	4	26.02	-1.93×10^4	128.0	80	0.32
Strang-SVGD	4	30.09	-1.20×10^5	121.7	121	1.55

Table 1: 2D targets quality-cost summary across particle methods. Entries are medians over the 25 target-seed runs, using the best checkpoint selected by the KSD monitor for each run. "Fails" counts runs whose KSD monitor never attained a finite value; methods are ordered first by failure count and then by median KSD. KSD is the primary distributional quality metric; mean log probability is reported as a secondary score, ESS as a supplemental mixing indicator, and gradient evaluations / wall time as cost metrics. Moment errors and grid L_1 remain diagnostic quantities and are omitted from this main summary.

SVGD each fail on four of the five funnel seeds, so finite-run medians alone would understate their instability; this is why table 1 reports failure counts alongside the quality and cost summaries. Overall, these 2D results sharpen the main qualitative claim: adaptive multirate integration is most valuable when the interaction between drift and repulsion becomes strongly geometry-dependent, rather than as a uniform improvement on every benign target. In this setting, ESS is much less discriminative than KSD and mean log-density, so we treat it as a supplemental mixing indicator rather than a primary quality metric.

3.5. Mixture2D (mix8)

This benchmark emphasizes multimodal exploration and mass allocation across modes. All methods start from the same small Gaussian centered at the origin, away from the modal ring, and then run for a fixed budget of 1,000 iterations with checkpointing every 20 steps across 5 seeds. For this benchmark we use a shared multiscale RBF kernel with bandwidth factors $\{0.5, 1, 2\}$ for all particle methods so that local spacing and longer-range repulsion are both represented on the modal ring. We treat this as an experimental kernel-choice ablation rather than a change to the multirate update itself; prior SVGD work has also considered broader kernel families beyond a single scalar RBF [5, 20]. We define mode coverage by assigning each sample to its nearest mixture center and counting a mode as covered once it carries at least 5% of the empirical mass. The primary metrics are mode coverage, normalized mode entropy, and mode imbalance; KSD and mean log probability are retained as secondary diagnostics, with ESS reported as

Table 2: Mixture2D (mix8) mean final metrics over five seeds at the fixed 1,000-iteration budget. Higher is better for coverage and entropy; lower is better for KSD. We omit minimum mass because it is zero for all methods at this budget.

Method	Coverage↑	Entropy↑	KSD↓
Adapt-MR-SVGD	0.500	0.544	0.360
SVGD	0.375	0.453	3162.9
Strang-SVGD	0.325	0.403	3883.0
MR-SVGD	0.250	0.238	74.3
SGHMC	0.125	0.000	2.29
SGLD	0.125	0.000	2.53

a supplemental mixing proxy. Here imbalance is computed from the same nearest-center empirical mode masses and measures deviation from uniform mass allocation across the eight modes.

Under this harder off-mode initialization, adaptive MR-SVGD still gives the strongest overall multimodal exploration at the fixed budget, now reaching the highest mean final mode coverage and mode entropy across seeds (0.500 and 0.544, respectively); table 2 provides the corresponding mean-over-seeds summary, while fig. 5 carries the cross-seed variability. The shared multiscale kernel also makes the single-rate particle baselines more competitive than under the plain single-RBF setting: vanilla SVGD and Strang-SVGD both cover more modes and attain higher mode entropy, but their KSD values remain far worse than the adaptive method, so the extra spread still comes with much weaker stability. Fixed MR-SVGD remains stable and slightly improves its KSD over the single-RBF version, but it does not match the adaptive variant on the main multimodal metrics. The imbalance panel in fig. 5 complements coverage and entropy by showing that the adaptive method also allocates mass more evenly across the discovered modes. SGLD and SGHMC remain trapped in a single mode, with final coverage fixed at 0.125 and zero mode entropy. The minimum mass statistic is still zero for all methods at this budget, so in this redesigned benchmark coverage, entropy, and imbalance are the informative multimodal diagnostics while KSD remains a secondary stability measure.

3.6. UCI logistic regression

We benchmark Bayesian logistic regression on breast_cancer, ionosphere, spambase, and a5a, using a Gaussian prior on weights (as implemented in

Method	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
breast_cancer			
Adapt-MR-SVGD	0.062	0.986	8.9
MR-SVGD	0.089	0.981	6.2
Strang-SVGD	0.094	0.979	6.5
SVGD	0.104	0.979	2.0
ionosphere			
Adapt-MR-SVGD	0.383	0.900	3.4
SVGD	0.438	0.906	0.9
Strang-SVGD	0.447	0.897	5.2
MR-SVGD	0.500	0.897	6.1
spambase			
SVGD	0.288	0.918	1.3
Strang-SVGD	0.288	0.924	8.8
Adapt-MR-SVGD	0.361	0.882	8.1
MR-SVGD	0.467	0.877	4.3
a5a			
Adapt-MR-SVGD	0.453	0.778	36.5
Strang-SVGD	0.471	0.836	3.0
MR-SVGD	0.474	0.809	14.6
SVGD	0.478	0.835	2.6

Table 3: UCI logistic-regression particle-method summary over five seeds. For each dataset and method, we select the best finite-NLL checkpoint per seed and report the method-wise mean of that best NLL, the accuracy at that checkpoint, and the corresponding wall time. ECE is summarized in the per-dataset panels of fig. 6.

the benchmark script) and 5 seeds per dataset. We report test accuracy, NLL, ECE, ESS, and mean log probability; KSD is optional and disabled by default in this benchmark configuration. Results are summarized with per-dataset panels in fig. 6 and a grouped best-NLL summary table (particle methods only) in table 3; both use the best finite-NLL checkpoint selected under the global predictive stopping policy [21].

UCI logistic regression serves mainly as a predictive sanity check rather than as the flagship evidence for multirate gains. The rankings are mixed. Adapt-MR-SVGD gives the best overall NLL on breast_cancer, while on ionosphere, spambase, and a5a either SGHMC or a single-rate particle method is competitive or superior. This weaker separation is consistent with the main claim of the paper: multirate benefits are most pronounced on anisotropic, multimodal, or hierarchical posteriors, whereas lower-dimensional logistic regression tasks show smaller and less uniform gains. On the smaller datasets, accuracy is also tightly quantized, so NLL is the more discriminative summary.

Method	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
spambase			
Adapt-MR-SVGD	0.174	0.940	35.7
Strang-SVGD	0.187	0.930	51.2
SVGD	0.187	0.930	21.1
MR-SVGD	0.188	0.931	51.5
a5a			
MR-SVGD	0.334	0.844	5.3
Adapt-MR-SVGD	0.347	0.836	8.1
SVGD	0.362	0.834	17.5
Strang-SVGD	0.378	0.832	49.6

Table 4: BNN particle-method summary over five seeds. For each dataset and method, we select the best finite-validation-NLL checkpoint per seed and report the method-wise mean of the corresponding test NLL, test accuracy, and wall time. ECE is summarized in the per-dataset panels of fig. 7.

3.7. Bayesian neural network

We evaluate a one-hidden-layer BNN (hidden width 32) on spambase and a5a, using a Gaussian prior, a fixed split seed, and 5 sampler seeds per dataset. For each dataset, we carve a validation split from the training data, select checkpoints by validation NLL under the global predictive stopping policy, and report test accuracy, test NLL, test ECE, ESS, and mean log probability at the selected checkpoint. KSD is optional and disabled by default in this configuration.

The two BNN datasets give a more informative but less uniform predictive story than the earlier small-data configuration. On spambase, Adapt-MR-SVGD attains the best overall test NLL (0.174) and the highest mean test accuracy (0.940), with the remaining particle methods clustered together around 0.187. On a5a, the ranking shifts: MR-SVGD attains the best overall test NLL (0.334) and the highest mean test accuracy (0.844), Adapt-MR-SVGD remains the second-best particle method (0.347), and the single-rate particle variants are again worse. The chain baselines are competitive in isolated cases—SGHMC on spambase and SGLD on a5a—but the multirate particle methods form the strongest pair overall across the two datasets. This makes the BNN benchmark corroborative rather than primary evidence: multirate structure improves predictive quality over the single-rate particle baselines, but the preferred multirate variant depends on the dataset.

3.8. Large-scale hierarchical logistic regression

To stress-test multirate behavior in a high-dimensional anisotropic posterior, we use a large-scale hierarchical logistic regression benchmark with grouped random effects. The model combines high parameter dimension with funnel-like geometry induced by a global scale parameter, making it a natural benchmark for joint quality-cost assessment of SVGD variants.

Problem definition. Given binary outcomes $y_i \in \{0, 1\}$, sparse features $x_i \in \mathbb{R}^p$, and group indices $g_i \in \{1, \ldots, G\}$, we define

$$\begin{gathered} y_i \mid x_i, g_i, \beta, \alpha, u \sim \mathrm{Bernoulli}\big(\sigma\big(\alpha + x_i^\top \beta + u_{g_i}\big)\big)\,, \\ i = 1, \ldots, n, \end{gathered} \tag{22}$$

where $\sigma(\cdot)$ is the logistic link. We use a non-centered parameterization for group effects,

$$\begin{aligned} \beta &\sim \mathcal{N}(0, \sigma_\beta^2 I_p), & \alpha &\sim \mathcal{N}(0, \sigma_\alpha^2), \\ u_j &= \tau z_j, & z_j &\sim \mathcal{N}(0, 1), \\ \log \tau &\sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \end{aligned} \tag{23}$$

for $j = 1, \ldots, G$. The posterior (up to normalization) is

$$\begin{aligned} \pi(\theta) \propto \Bigg[\prod_{i=1}^{n} \mathrm{Bernoulli}\big(y_i; \sigma\big(\alpha + x_i^\top \beta + u_{g_i}\big)\big)\Bigg] \\ \times\, p(\beta)\, p(\alpha)\, p(z)\, p(\log \tau), \\ \theta = (\beta, \alpha, z, \log \tau). \end{aligned} \tag{24}$$

Protocol. For the long-tail grouped setting, we run $n = 10^6$ observations with $p = 300$ features and $G = 50{,}000$ groups. Each method runs up to 1,000 iterations with checkpointing every 20 iterations and 5 random seeds; particle methods use 32 particles, and single-chain baselines use a rolling-window evaluation set. This benchmark follows the global predictive NLL-based stopping policy and adds a two-checkpoint non-finite fail-fast guard because unstable methods can quickly leave the finite-NLL regime. ECE is reported as a metric but is not used as a stopping trigger.

Result summary. table 5 shows that Adapt-MR-SVGD achieves the best mean best-NLL while remaining finite across all seeds. MR-SVGD is a close second in NLL and reaches its best checkpoint faster on average (92.5 s versus 123.3 s). SVGD/Strang-SVGD and SGHMC show frequent non-finite

Method	Finite-best↑	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
Adapt-MR-SVGD	5/5	0.613	0.658	123.3
MR-SVGD	5/5	0.623	0.656	92.5
SGHMC	2/5	0.860	0.560	77.9
SVGD	3/5	0.885	0.577	154.2
Strang-SVGD	3/5	1.263	0.579	163.6
SGLD	4/5	1.823	0.576	101.6

Table 5: HLR long-tail summary over five seeds. For each seed and method, we select the best finite-NLL checkpoint; the table reports method-wise means of that best NLL, the accuracy at that checkpoint, and the corresponding wall time. "Seeds with finite best" counts runs that reached at least one finite-NLL checkpoint before stopping.

Method	Finite-best↑	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
Adapt-MR-SVGD	5/5	0.681	0.601	94.6
SVGD	1/5	0.700	0.548	54.9
MR-SVGD	5/5	0.706	0.597	44.2
Strang-SVGD	1/5	0.708	0.584	241.7
SGLD	5/5	0.743	0.528	66.0
SGHMC	5/5	0.791	0.554	79.7

Table 6: HLR uniform-group summary over five seeds. As in table 5, each entry uses the best finite-NLL checkpoint per seed and reports method-wise means for NLL, accuracy at that checkpoint, and wall time.

behavior in this regime, while SGLD remains finite more often but with substantially worse NLL.

Uniform-group sensitivity check. table 6 provides a compact robustness check under uniform group assignments. Adapt-MR-SVGD remains best in mean best-NLL with full finite-seed coverage (5/5), and MR-SVGD remains close while being fastest among methods with full finite coverage. This supports the same ordering seen in the long-tail regime while keeping the long-tail setting as the primary stress test for our main claim.

4. Conclusions

This work reframes SVGD as a numerically split interacting-particle flow in which attraction to high-density regions and diversity-preserving repulsion are treated as distinct computational components. From this viewpoint, multirate integration is not only an efficiency heuristic but a practical mechanism for improving stability in posterior geometries where these components evolve at different effective scales.

Across a broad benchmark suite, the main empirical pattern is that multirate formulations make SVGD behavior more reliable under difficult geometry, especially in high-dimensional anisotropic settings and hierarchical models. Fixed multirate schedules provide a strong robustness baseline, while adaptive drift resolution offers additional flexibility when local stiffness changes along the trajectory. At the same time, results indicate that no single variant dominates every metric and every regime, reinforcing the importance of evaluating quality and cost jointly rather than through a single summary statistic.

Several limitations remain. Current adaptivity is centered on drift resolution, while repulsion frequency is still controlled by simple schedules; ESS remains a univariate proxy; and dense kernel interactions continue to constrain scaling at large particle counts. These observations suggest clear next steps: tighter adaptive controllers that coordinate drift and repulsion updates, approximate or structured kernel computations for large-N regimes, and stronger theoretical links between multirate stability analysis and particle-based Bayesian inference guarantees. Together, these directions can move multirate SVGD from a robust empirical strategy toward a more principled and scalable foundation for modern Bayesian computation.

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References

[1] M. Welling, Y. W. Teh, Bayesian learning via stochastic gradient langevin dynamics, in: International Conference on Machine Learning, 2011.

[2] T. Chen, E. Fox, C. Guestrin, Stochastic gradient hamiltonian monte carlo, in: International Conference on Machine Learning, 2014.

[3] C. Liu, J. Zhuo, P. Cheng, R. Zhang, J. Zhu, Understanding and accelerating particle-based variational inference, in: K. Chaudhuri, R. Salakhutdinov (Eds.), Proceedings of the 36th International Conference on Machine Learning, Vol. 97 of Proceedings of Machine Learning Research, PMLR, 2019, pp. 4082–4092.
URL `https://proceedings.mlr.press/v97/liu19i.html`

[4] J. Zhuo, C. Liu, J. Shi, J. Zhu, N. Chen, B. Zhang, Message passing stein variational gradient descent, in: J. Dy, A. Krause (Eds.), Proceedings of the 35th International Conference on Machine Learning, Vol. 80 of Proceedings of Machine Learning Research, PMLR, 2018, pp. 6018–6027.
URL `https://proceedings.mlr.press/v80/zhuo18a.html`

[5] D. Wang, Z. Tang, C. Bajaj, Q. Liu, Stein variational gradient descent with matrix-valued kernels, in: Advances in Neural Information Processing Systems, 2019.

[6] Q. Liu, D. Wang, Stein variational gradient descent: A general purpose bayesian inference algorithm, in: Advances in Neural Information Processing Systems, 2016.

[7] J. Zhang, R. Zhang, L. Carin, C. Chen, Stochastic particle-optimization sampling and the non-asymptotic convergence theory, in: S. Chiappa, R. Calandra (Eds.), Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics, Vol. 108 of Proceedings of Machine Learning Research, PMLR, 2020, pp. 1877–1887.
URL `https://proceedings.mlr.press/v108/zhang20d.html`

[8] L. Pareschi, G. Russo, Implicit-explicit Runge-Kutta schemes for stiff systems of differential equations, in: Recent trends in numerical analysis, Nova Science Publishers, Inc., 2000, pp. 269–288.

[9] A. Sarshar, S. Roberts, A. Sandu, Parallel implicit-explicit general linear methods, Communications on Applied Mathematics and Computation 3 (2021) 649–669. doi:10.1007/s42967-020-00083-5.

[10] M. Günther, A. Sandu, Multirate generalized additive Runge-Kutta methods, Numerische Mathematik 133 (3) (2016) 497–524.

doi:10.1007/s00211-015-0756-z.
URL `http://dx.doi.org/10.1007/s00211-015-0756-z`

[11] A. Sarshar, S. Roberts, A. Sandu, Design of high-order decoupled multirate gark schemes, SIAM Journal on Scientific Computing 41 (2) (2019) A816–A847. doi:10.1137/18M1182875.

[12] A. Sandu, M. Günther, S. Roberts, A. Sarshar, Implicit multirate gark methods, Journal of Scientific Computing 87 (2021) 1–32. doi:10.1007/s10915-020-01400-z.

[13] A. Sarshar, S. Roberts, A. Sandu, Alternating directions implicit integration in a general linear method framework, Journal of Computational and Applied Mathematics 387 (2021) 112496. doi:10.1016/j.cam.2019.112496.

[14] A. Sarshar, S. Roberts, A. Sandu, A fast time-stepping strategy for dynamical systems equipped with a surrogate model, SIAM Journal on Scientific Computing 44 (3) (2022) A1405–A1430. doi:10.1137/20M1386281.

[15] G. Strang, On the construction and comparison of difference schemes, SIAM Journal on Numerical Analysis 5 (3) (1968) 506–517.

[16] E. Hairer, S. Norsett, G. Wanner, Solving ordinary differential equations I: Nonstiff problems, no. 8 in Springer Series in Computational Mathematics, Springer-Verlag Berlin Heidelberg, 1993. doi:10.1007/978-3-540-78862-1.

[17] Q. Liu, J. Lee, M. Jordan, A kernelized stein discrepancy for goodness-of-fit tests, in: International Conference on Machine Learning, 2016.

[18] J. Gorham, L. Mackey, Measuring sample quality with kernels, in: D. Precup, Y. W. Teh (Eds.), Proceedings of the 34th International Conference on Machine Learning, Vol. 70 of Proceedings of Machine Learning Research, PMLR, 2017, pp. 1292–1301.
URL `https://proceedings.mlr.press/v70/gorham17a.html`

[19] T. Gneiting, A. E. Raftery, Strictly proper scoring rules, prediction, and estimation, Journal of the American Statistical Association 102 (477) (2007) 359–378.

[20] A. S. Stordal, R. J. Moraes, P. N. Raanes, G. Evensen, p-kernel stein variational gradient descent for data assimilation and history matching, Mathematical Geosciences 53 (2021) 375–393. doi:10.1007/s11004-021-09937-x.

[21] D. Dua, C. Graff, Uci machine learning repository, `https://archive.ics.uci.edu/ml`, accessed 2025-01-01 (2019).

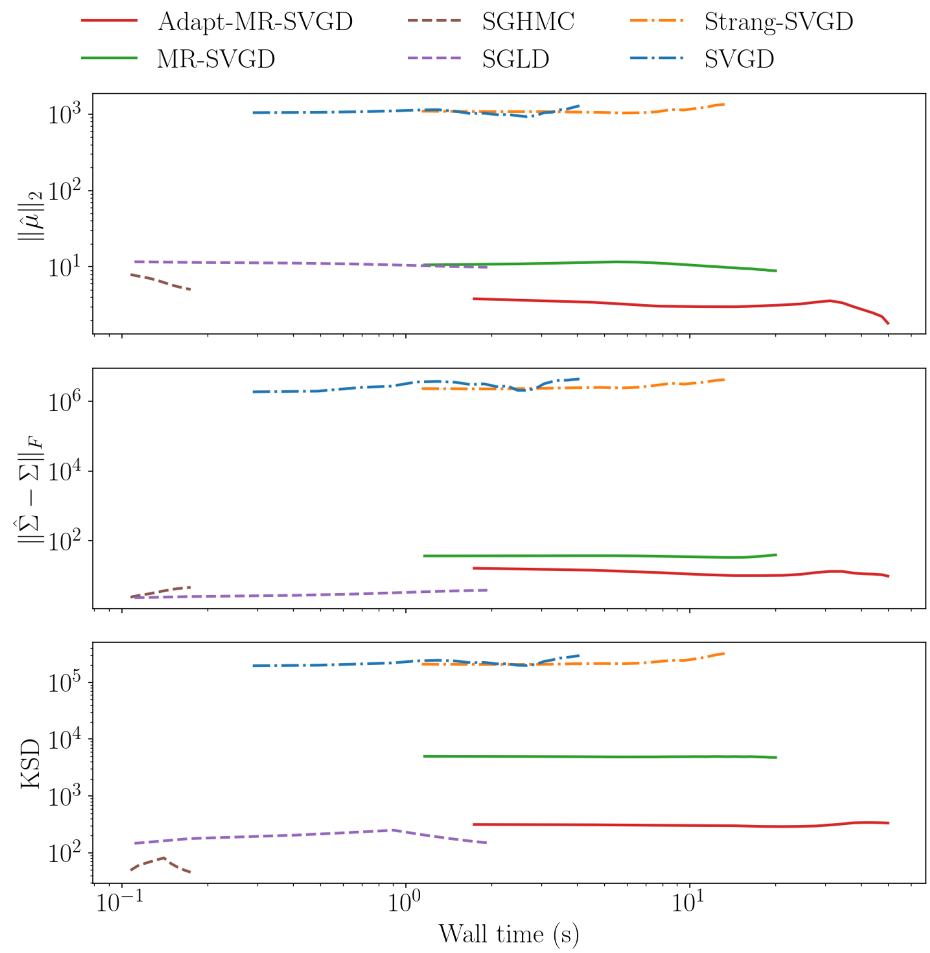
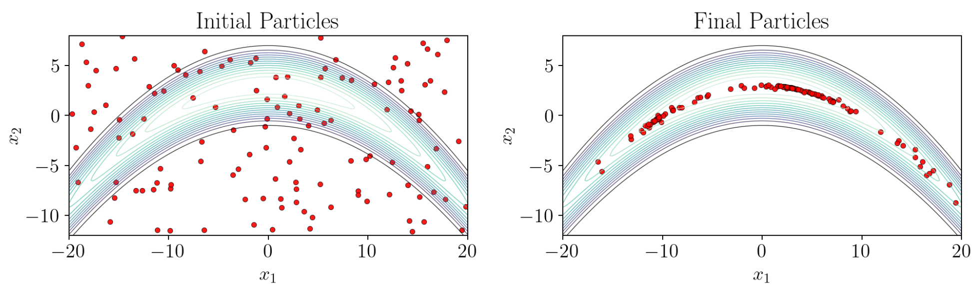
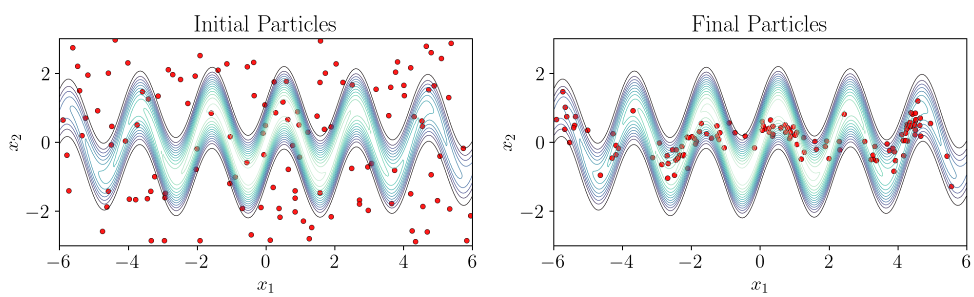


Figure 3: 50D Gaussian: wall-time overview across key metrics.

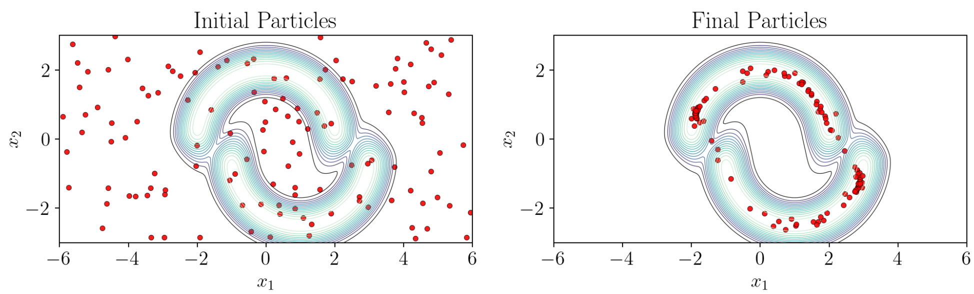


(a) Banana target: initial and short-run final particle states.

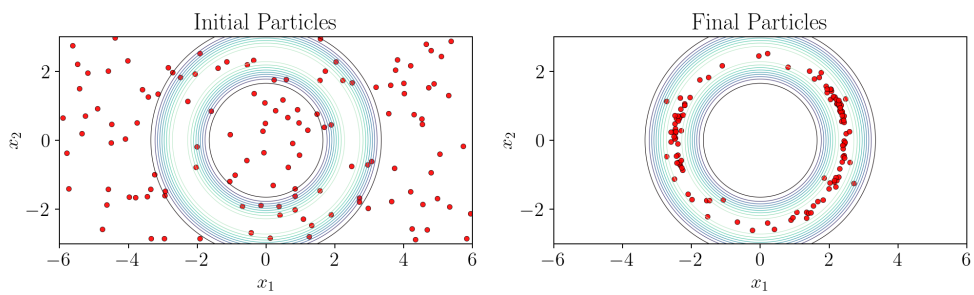


(b) Squiggly target: initial and short-run final particle states.

Figure 4: 2D target visualization panels produced with Adapt-MR-SVGD under visualization-only settings. Each subfigure shows target-density contours with initial particles (left) and short-run final particles (right).



(c) Two-moons target: initial and short-run final particle states.



(d) Ring target: initial and short-run final particle states.

Figure 4: 2D target visualization panels (continued).

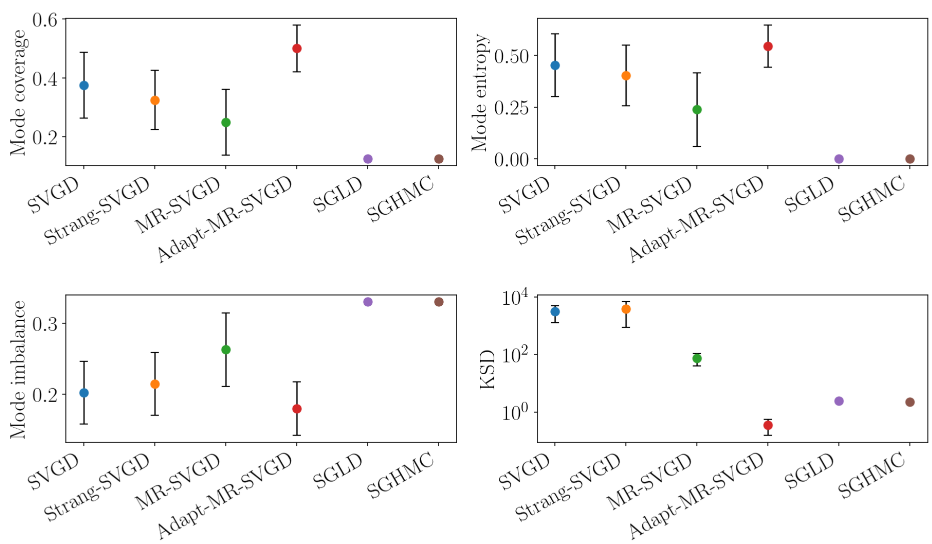
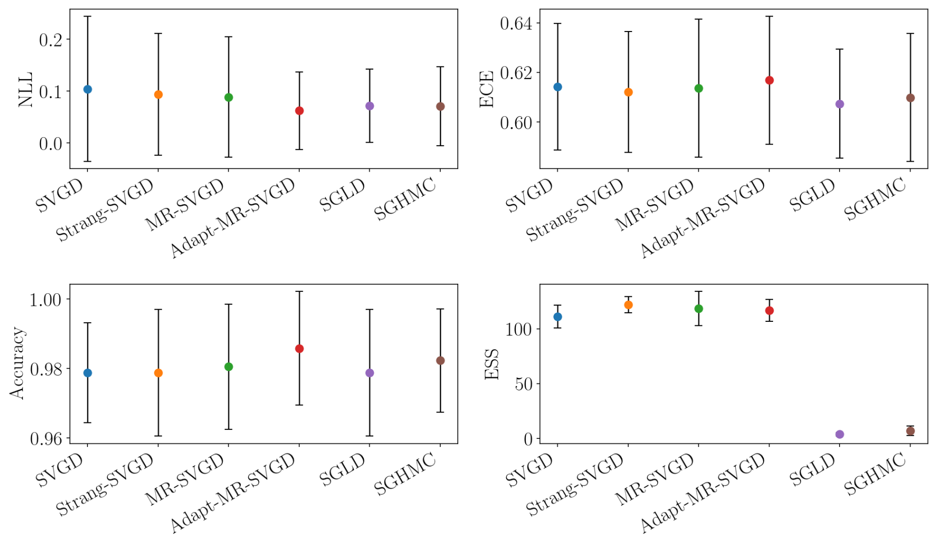
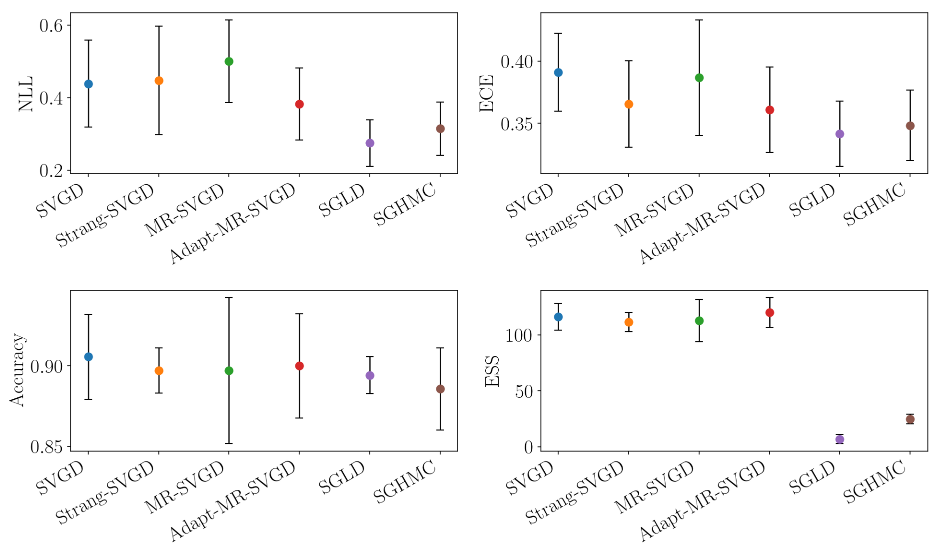


Figure 5: Mixture2D (mix8) fixed-budget final-checkpoint summary across methods. The four panels report mode coverage, mode entropy, mode imbalance, and KSD; markers show mean values across seeds and error bars indicate one standard deviation. Higher is better for coverage and entropy, while lower is better for mode imbalance and KSD.

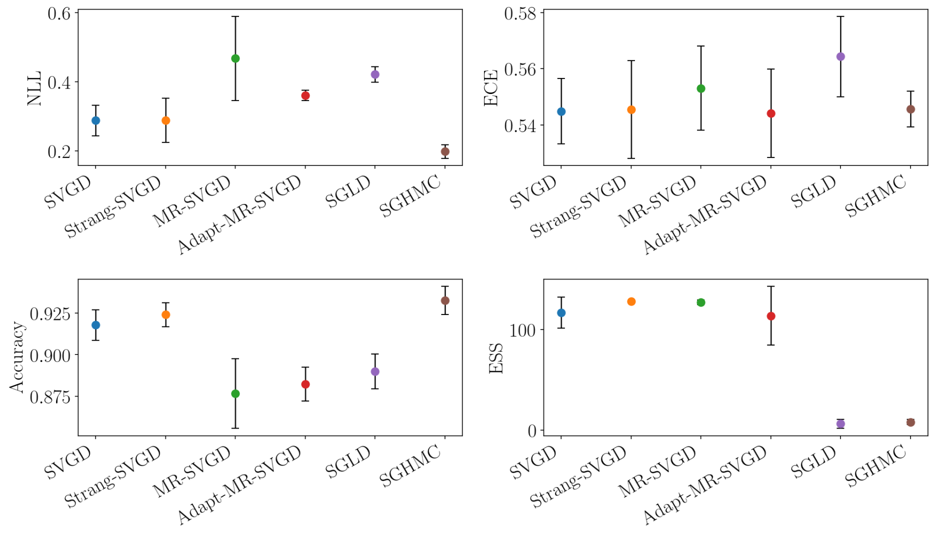


(a) breast_cancer

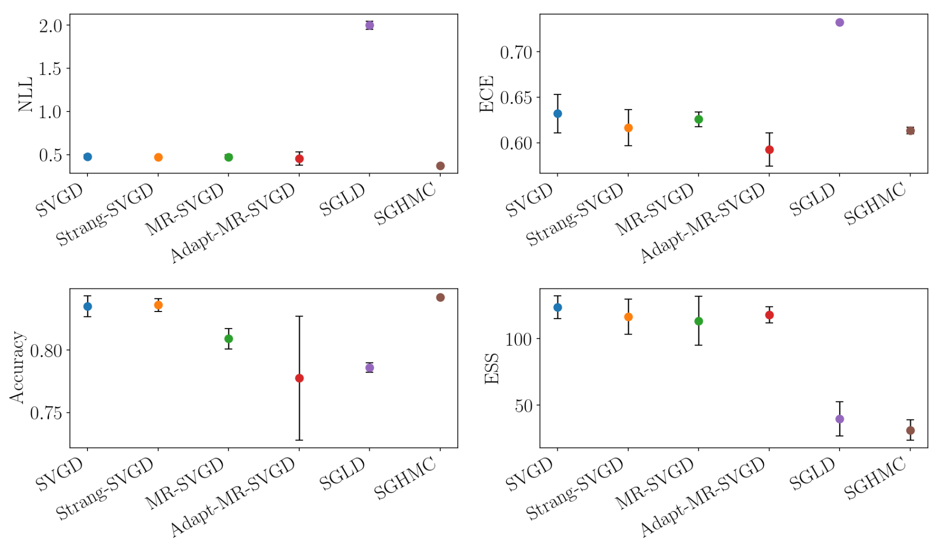


(b) ionosphere

Figure 6: UCI logistic regression summary across datasets. Each subpanel reports the best-checkpoint mean across seeds with one standard deviation for test accuracy, NLL, ECE, and ESS, where the checkpoint is selected by the NLL-based early-stop monitor. Higher is better for accuracy and ESS; lower is better for NLL and ECE. Remaining datasets are shown in the continued panel.

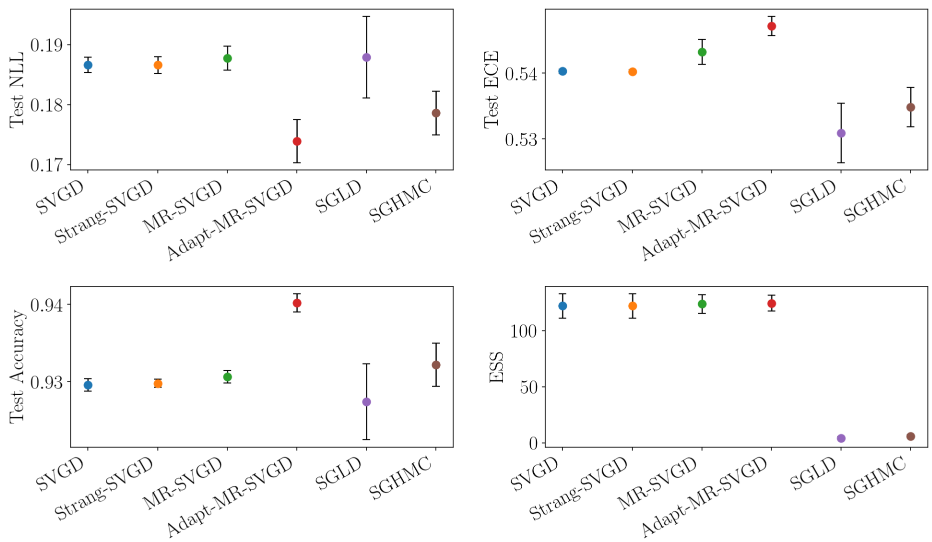


(c) spambase

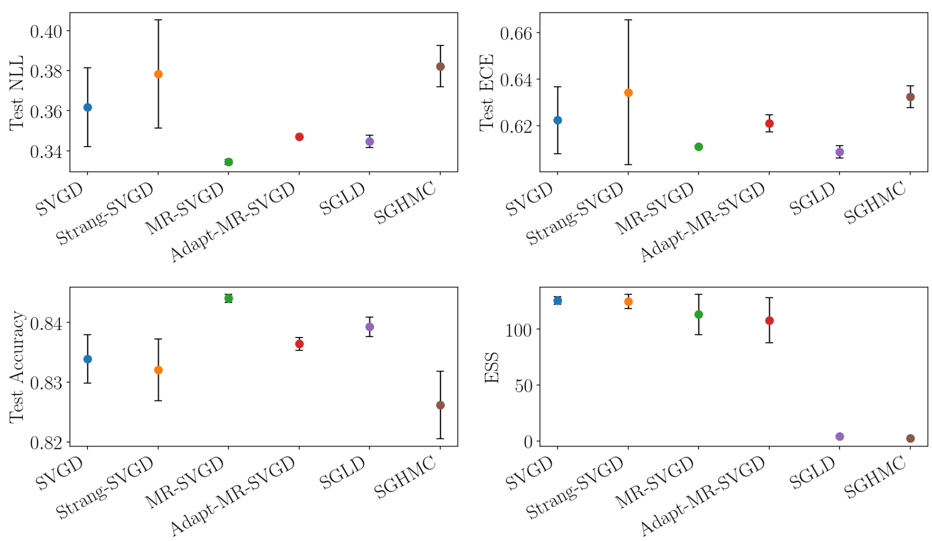


(d) a5a

Figure 6: UCI logistic regression summary across datasets (continued).



(a) spambase



(b) a5a

Figure 7: BNN summary across datasets. Each subpanel reports the best-checkpoint mean across seeds with one standard deviation for test accuracy, NLL, ECE, and ESS, where the checkpoint is selected by the validation-NLL early-stop monitor. Higher is better for accuracy and ESS; lower is better for NLL and ECE.